LESSON PLAN

AVIONICS AND ELECTRONIC WARFARE (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	17
Total No. of Lesson Plans	25	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Purpose & operation of A/C external lighting: - Navigation lights, anti-collision lights, landing lights and taxing lights, Ice inspection lights
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Purpose & operation of A/C external lighting: - Navigation lights, anti-collision lights, landing lights and taxing lights, Ice inspection lights practical on position and wiring of lights.
5. Specific Objectives	13.1.3.1 Knowledge on circuit protection devices, generators, actuators & engine starting system.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Practical on position and wiring of external lights; Safety: Verify power off before touching wiring; Supervision: Instructor monitors connections and circuit protection verification.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of navigation lights?
(b) Which color is associated with the port side navigation light?
(c) What is the purpose of anti-collision lights?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Navigation lights color and position • Anti-collision light operation types • Landing light circuit and control • Taxi light activation and location • Ice inspection light usage	L & Ds	PPT, IB	15 min
(b)	• Circuit protection devices for lighting • Wiring harness identification and routing • Switch operation and signal flow • Grounding points and continuity checks • Fuses and relays specific to lights	L & Ds	PPT, PDF	20 min
(c)	• Troubleshooting light failure scenarios • Verification of correct light rotation • Inspection of bulb integrity and sockets • Testing circuit protection reset functions • Documentation of light system status	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between flashing and steady anti-collision lights.
(b) Describe the circuit path for the landing light from the switch to the bulb.
(c) How do you verify the correct installation of navigation lights?

12. Home Assignment and Follow Up	Prepare a brief note on the basic circuit of the turbo prop engine starting system to be discussed in LP 18.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Introduction to turbine engine starting: - Basic circuit of turbo prop engine starting system, turbo starter system, and starter generator system. Safeties to be taken while gas turbine engine starting. Practical on engine starting system.	Refer to the master syllabus reference list	LP 18

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min